Set 11 at Daan — the front window where the cab would be.

← Back to the previous page

VAL256

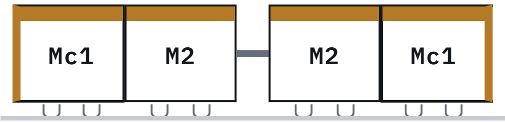
The French fleet that opened the network in 1996 — and spent eighteen months in a depot having its brain replaced while the line it built ran without it.

BR WENHU LINE [/RAIL/LINES/WENHU-LINE/] 文湖線

BUILDER	FAMILY	OPERATOR	LINE
Matra Transport, with GEC Alsthom [1]	VAL	Taipei Rapid Transit Corporation [/rail/operators/trtc/]	Wenhu [/rail/lines/wenhu-line/]
BUILT	**IN SERVICE**	**DEPOT**	
1990—1993 [1]	28 March 1996 — [6]	Muzha [/rail/depots/muzha-depot/] and Neihu [/rail/depots/neihu-depot/] [7]	

Stub page. This entry is an outline. Figures marked *TBC* still need to be checked against a primary source before publication.

Formation



4 CARS · 2 UNITS OF 2 · ONE UNIT COUPLING CAB

A train delivered in the early 1990s is still running the line, and almost nothing tells it what to do any more.

The VAL256 sets were built by Matra with GEC Alsthom between 1990 and 1993[1] and entered service with the Muzha Line on 28 March 1996[6]. The `256` is the car width in centimetres — 2,560 mm — and it is the wider of the two standard VAL profiles, the other being the 2,060 mm VAL206 used at Lille[4].

The re-signalling

The conversion did not precede the Neihu [/rail/stations/br19/] extension — it was triggered by it, and the sequence is the story:

DATE	EVENT
4 July 2009	The Neihu extension opens. The line switches to CITYFLO 650, the Matra system is cut off after thirteen years, and the VAL256 fleet **comes out of service that day** to begin conversion[1] [5]
26 December 2010	Converted VAL256 units return to passenger service, running alongside the Innovia fleet[1]

So for nearly eighteen months the whole 24-station line ran on the Bombardier fleet alone, while the trains that had opened the railway sat in a depot having their control system replaced. Why the line was re-signalled at all — and why by a competitor of the system's owner — is told on The Matra dispute [/rail/history/matra-dispute/].

The original equipment was a fixed-block ATC/ATO system derived from the MAGGALY technology on Lyon Metro line D[2]; what replaced it is a moving-block communications-based system.

Design

Fleet history

Depot allocation

Corrections

Specifications

Car width	2.56 m[1]
Car length	13.78 m[1]
Car height	3.53 m[1]
Cars per train	4 [1]
Empty weight	TBC t
Design maximum speed	80 km/h[1]
Operating speed	70 km/h[1]
Seated capacity	20 per car (AW2) [1]
Standing capacity	94 per car (AW2) [1]
Train capacity	440 [11]
Doors per car	4 2 per side[1]
Power supply	750 V DC[1]
Traction	GEC Alsthom chopper DC[1]
Motors per car	2 [1]
Braking	Regenerative and disc [1]
Guidance	Side guide bars [3]
Signalling	CITYFLO 650 since 2010[1]
Automation	GoA4 (UTO) [5]
Fleet size	102 cars[1]
Pairs	51 [8]
Tyre replacement interval	TBC

References

3 of 11 sources on this page are primary — published by the operator, the builder or government.

1 Taipei Metro VAL256 electric multiple unit [https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]

台北捷運VAL256型電聯車
SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06
archived [https://web.archive.org/web/20221006051618/https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]
The fullest specification found for this fleet. The article carries no reference section, so every figure taken from it is one source deep — corroborated below by the English article, which is independently worded.

2 Taipei Metro VAL256 [https://en.wikipedia.org/wiki/Taipei_Metro_VAL256]
SECONDARY English Wikipedia accessed 2026-08-06
archived [https://web.archive.org/web/20260511224742/https://en.wikipedia.org/wiki/Taipei_Metro_VAL256]
Agrees with the Chinese article on every dimension, capacity, speed and date. Adds that the original signalling derived from the MAGGALY system on Lyon Metro line D.

3 Lille VAL — project profile [https://www.railway-technology.com/projects/lille_val/]
SECONDARY Railway Technology accessed 2026-08-06
archived [https://web.archive.org/web/20260807044500/https://www.railway-technology.com/projects/lille_val/]
Describes the guideway directly — running surfaces, lateral H-section guide bars about 200 mm above the running surface, and those bars doubling as the 750 V supply. Trade press, not a manufacturer drawing; a Siemens or Matra engineering document would settle it properly.

4 Véhicule Automatique Léger [https://en.wikipedia.org/wiki/V%C3%A9hicule_Automatique_L%C3%A9ger]
SECONDARY English Wikipedia accessed 2026-08-06
archived [https://web.archive.org/web/20260807045415/https://en.wikipedia.org/wiki/V%C3%A9hicule_Automatique_L%C3%A9ger]
Independent corroboration of side guidance — "pairs of horizontal tires to provide lateral guidance", with power "collected by shoes from the guidebars". Also records that the later NeoVal moved to a single central rail, which is the most likely origin of the central-rail error this page used to carry.

5 Wenhu line [https://zh.wikipedia.org/zh-tw/文湖線]
文湖線
SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06
archived [https://web.archive.org/web/20250714045049/https://zh.wikipedia.org/zh-tw/文湖線]

6 Wenshan—Neihu Line [https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94]
文山內湖線
PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS) accessed 2026-08-06
archived [https://web.archive.org/web/20260209210832/https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94]

7 Muzha Depot [https://zh.wikipedia.org/zh-tw/木柵機廠]
木柵機廠
SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06
archived [https://web.archive.org/web/20220520160336/https://zh.wikipedia.org/zh-tw/木柵機廠]

8 How many trains were procured for each line of the Taipei metro network? [https://www.dorts.gov.taipei/News_Content.aspx?n=2A66A485FACB0D5B&s=C8602F8588914E91]
台北都會區大眾捷運系統路網中，各路線所採購之列車數為何？
PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS) accessed 2026-08-07
archived [https://web.archive.org/web/20260807044541/https://www.dorts.gov.taipei/News_Content.aspx?n=2A66A485FACB0D5B&s=C8602F8588914E91]
The builder's own procurement table: 木柵線 CC350 51對電聯車（102輛）. The contract number and the pair count.

9 Taipei Metro rolling stock [https://zh.wikipedia.org/zh-hant/臺北捷運列車]
臺北捷運列車
SECONDARY 維基百科（Chinese Wikipedia） accessed 2026-08-07
archived [https://web.archive.org/web/20260807045301/https://zh.wikipedia.org/zh-hant/%E8%87%BA%E5%8C%97%E6%8D%B7%E9%81%8B%E5%88%97%E8%BB%8A]
The only source found for this fleet running 「非固定編組」 — a non-fixed formation, four consecutively-numbered cars making a train rather than a permanent four-car unit. It is the claim the site's formation notation used to contradict, and no manufacturer or operator document has been found either way.

10 Taipei Metro turns space magician: after the Bombardier fleet, the Wenhu line's Matra trains get an interior optimisation [https://www.metro.taipei/News_Content.aspx?n=30CCEFD2A45592BF&sms=72544237BBE4C5F6&s=86F98235BC105C44]
臺北捷運化身空間魔術師！繼龐巴迪列車改善後 文湖線馬特拉列車空間優化
PRIMARY Taipei Rapid Transit Corporation (news release) accessed 2026-08-06
archived [https://web.archive.org/web/20250723182554/https://www.metro.taipei/News_Content.aspx?n=30CCEFD2A45592BF&sms=72544237BBE4C5F6&s=86F98235BC105C44]
The operator's own announcement of the seat-removal programme, and the source for this fleet forming 25 four-car trains.

11 The whole Wenhu line to be renewed — new trains gangwayed across four cars, capacity up 30% to 574 [https://www.ettoday.net/news/20260507/3162180.htm]
文湖線將全線更新　新列車採「四節貫穿」、載客量增30％至574人
SECONDARY ETtoday 新聞雲 , 7 May 2026 accessed 2026-08-06
archived [https://web.archive.org/web/20260807045443/https://www.ettoday.net/news/20260507/3162180.htm]
Gives 440 as the current Matra train capacity, as the comparison for the replacement programme. Treat as provisional — it is a figure quoted in a council exchange reported in news, not a published specification, and it does not reconcile exactly with 114 per car.

Last updated: 2026-08-06

Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operators TRTC, NTMC, and TYMC. Government open data. Retrieved 2026-08-10; source last updated 2020-01-31. The 12 Sanying station records absent from TDX are sourced directly to New Taipei Metro and government publications on their pages. Provenance.